

Management

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**We agree that equal and fair contributions were made to the final
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**Please list brief details of how tasks were divided up among the group members in the
box below. (Please refer to chapters of final report etc.)**

2789745: Model Justification, Business Insights : finding 2
2678637: Business insights: finding 1, Conclusion, Limitation
2744101: Coding, Methodology, Results
2678619: Introduction, Data Description

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Banking Efficiency in the UK: A Comparison of Domestic and Foreign-Owned Banks

Date: 09.05.2026

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Introduction

UK Finance (2023) points out that the banking sector plays an important role in the UK economy, contributing over 8% of GDP, employing more than one million people, and supporting both business activity and household finance across the country. Early studies in the UK financial services industry often relied on cost function approaches that implicitly assume firms are highly efficient in operation, except for their inefficiency in size or scope (Drake, 2001). This study aims to examine how UK-listed and UK-headquartered banks effectively convert their resources into financial and lending outcomes, and see whether the domestically-owned and foreign-owned banks have different performances.

Performance metrics such as return on equity (ROE) and return on assets (ROA) are commonly used to evaluate bank performance. However, they may have potential limitations. For example, ROE does not take risk into account, indicating that higher returns can be brought by taking more risk. This may end up with a misleading conclusion (Klaassen and van Eeghen, 2015). Different characteristics from these ratios, like size and specialisation, also make it difficult to build comparable benchmarks, which limits the efficiency assessment (Maradin et al., 2018). Lastly, these metrics evaluate performance dimensions independently, without considering multiple inputs and outputs at the same time. This causes further challenges in identifying the specific reason for inefficiency. Therefore, the Data Envelopment Analysis (DEA) method was selected in this study to address the shortcomings. DEA allows assessing the relative efficiency between comparable units through incorporating more than one input and output, making it reliable and suitable for complex domains such as the banking industry. Additionally, it enables identifying sources of inefficiency and ranking the DMUs (Golany and Roll, 1989). The analysis will apply CRS and VRS DEA models, with super-efficiency ranking, to compare efficiency patterns across bank groups and years.

This study could benefit bank managers by reviewing how well resources are being used to maximise value. It may also help investors gain a clearer understanding of relative bank performance beyond simple financial ratios.

1. Data Description

Data was collected from six banks across the financial years 2022 to 2025. Each bank year was considered an individual Decision Making Unit (DMU), resulting in a total of 24 DMUs. We gathered the data from the Income statement and Balance sheet reported in published annual reports for all six banks. For HSBC, Barclays, and Santander, the UK segment data was specifically selected from their breakdowns to ensure all variables have the same base. While choosing the variables, we followed the theory of intermediation, where banks allocate resources to create financial value (Allen and Santomero, 1997). For the inputs, total deposits and operating costs were selected since they represent the aggregate of funding and necessary expenses to maintain the business, respectively. On the other hand, net loans and total revenue served as outputs to show the lending activity and overall financial performance. Together, these variables form a concise representation of banks' financial intermediation performance. Other variables, such as the number of employees or total equity, were considered at first, but they were excluded because of high correlation and limited comparability. With 2 inputs and 2 outputs, the rule of thumb is satisfied by our 24 DMUs.

In terms of the potential data quality issues, the differences in accounting practices and the use of segment data from six banks may pose a risk. Therefore, this should be taken into account when interpreting the results.

2. Model Justification

The banks in our analysis notably differ in size, which could be justified by the number of employees and the total amount of deposits. For example, in 2025 Santander had around 16 thousand employees with 192 billion pounds in deposits, while Lloyds had 60 thousand employees and 496 billion total deposit amount, which is far more than double. Furthermore, banks' performance tends to be influenced by economies of scale, such that they benefit from the increased profits and reduced costs up to a certain size, and later they incur lower profits with the greater size (Stimpert and Laux, 2011). Therefore, given that the Constant Returns to Scale (CCR) DEA model measures overall efficiency under the assumption that a proportional increase in inputs leads to a proportional increase in outputs (Charnes et al., 1978), we may infer that the **VRS model** is more appropriate.

Hence, the Variable Returns to Scale (BCC) DEA model adds a convexity constraint that restricts each DMU to be benchmarked just for pure technical efficiency, removing scale effects, therefore, reflecting managerial efficiency independent of the bank size (Banker et al., 1984). However, we will still use the CRS scores to calculate further performance metrics such as **scale efficiency**.

As it was previously mentioned, one of the key objectives of this analysis is to investigate how effectively banks utilise their resources to maximise their financial performances. Therefore, if the main goal of the analysis is related to output expansion, then **output orientation** should take place (Cook et al., 2014). Consequently, the **envelopment programming formulation** will be used to calculate efficiency scores for each DMU.

Afterwards, the model outputs will be used to identify the best-performing banks and target output values for each inefficient bank. We can also consider the scale efficiency of each bank over 4 years and determine those that operate at the **most productive scale size**. Furthermore, it may also provide data-driven recommendations on whether the company's management is on the right track toward adjusting its operational scope (Dellnitz et al., 2018).

In our project, we aim to get a complete ranking of all banks in each year, and due to the fact that the traditional efficiency score is tied to a maximum score of one, we will apply the **super-efficiency technique**. Furthermore it will allow us to identify outliers that extend the efficient frontier and make other units' efficiencies overly optimistic (Błach and Wiecek-Kosmala, 2013).

3. Methodology

3.1 DEA Model

An output-oriented DEA model was implemented with the following variables for each DMU:

- X_{deposits} - total deposits (£m) (input)
- X_{opex} - operating expenses before credit impairment (£m) (input)
- Y_{loans} - net loans (£m) (output)

- Y_{revenue} - total revenue (£m) (output)

For each DMU j_0 , the model constructs a virtual composite unit as a convex combination of peer DMUs, weighted by the intensity weights (λ_k). The model then maximises the output expansion factor (ϕ) such that this composite produces at least ϕ times the outputs of DMU j_0 without exceeding its inputs.

Scale efficiency is calculated as the ratio between CRS and VRS efficiency. It measures whether a bank is operating at the most productive scale size, with a score of one indicating that the bank is operating at an optimal scale, while a score below one suggests scale inefficiency. Thus helping distinguish between operational inefficiency and inefficiency associated with organisational size.

$$\text{Scale Efficiency} = \frac{\text{CRS Efficiency}}{\text{VRS Efficiency}}$$

3.2 Super-Efficiency and Group Comparison

Super-efficiency excludes the evaluated DMU from its own reference set, allowing efficient banks to obtain a score greater than 1 and thereby enabling the full ranking of frontier units (Andersen and Petersen, 1993).

This study additionally compares domestic and foreign-owned banks separately by running separate DEA models for Lloyds, NatWest, and Nationwide, and for HSBC, Barclays UK, and Santander UK. Average CRS, VRS and scale efficiency scores are then compared between the two groups to examine whether ownership structure is associated with different efficiency patterns, with a Mann-Whitney U Test done to check for the statistical significance of this examination.

4. Results

Table 1: Input and output data for all six banks (2022-2025)

Bank	Year	Total Deposits (£m)	Operating expenses before credit impairment	Net loans (£m)	Total revenue (£m)
Santander UK	2022	197313	2370	223840	5006
	2023	195149	2485	211222	5176
	2024	185775	2577	202929	4683
	2025	192331	2494	205966	4796
Barclays UK	2022	258000	4327	205100	7259
	2023	241100	4415	202800	7587
	2024	244200	4329	207700	8274
	2025	244600	4882	216500	8708
Nationwide	2022	183175	2234	208066	3860
	2023	192334	2323	210782	4669
	2024	197896	2549	213440	4781
	2025	282095	3550	300889	7484
Lloyds	2022	475300	9759	454900	18209
	2023	471400	10823	449700	18629
	2024	494000	11601	459900	18003
	2025	496500	11966	481100	19422
NatWest	2022	450300	7687	366300	13156
	2023	431400	7996	381400	14752
	2024	433500	8149	400300	14703
	2025	443000	8262	418900	16641
HSBC	2022	291816	3832	210500	7952
	2023	279188	3707	219867	10807
	2024	335378	4310	224867	9979
	2025	326050	4017	237859	10468

Table 1 summarises input and output data for all six banks from 2022 to 2025. Lloyds and NatWest are notably larger in deposits and loans, while Nationwide and Santander UK operate at substantially smaller scales.

Table 2: DEA efficiency scores for all 24 DMUs

DMU	CRS	VRS	Scale Efficiency	Super-Efficiency	On Frontier
Barclays UK 2022	0.81	0.8305	0.9753	0.8305	No
Barclays UK 2023	0.8667	0.9159	0.9463	0.9159	No
Barclays UK 2024	0.9106	0.9607	0.9479	0.9607	No
Barclays UK 2025	0.9322	1	0.9322	1.0197	Yes
HSBC 2022	0.7942	0.8133	0.9765	0.8133	No
HSBC 2023	1	1	1	1.2305	Yes
HSBC 2024	0.8213	0.884	0.9291	0.884	No
HSBC 2025	0.927	0.9703	0.9554	0.9703	No
Lloyds 2022	1	1	1	1.0249	Yes
Lloyds 2023	1	1	1	1.0137	Yes
Lloyds 2024	0.9512	0.9646	0.9861	0.9646	No
Lloyds 2025	1	1	1	1.0588	Yes
NatWest 2022	0.832	0.9056	0.9187	0.9056	No
NatWest 2023	0.9238	0.9256	0.9981	0.9256	No
NatWest 2024	0.9394	0.9631	0.9754	0.9631	No
NatWest 2025	1	1	1	1.0345	Yes
Nationwide 2022	1	1	1	1.0367	Yes
Nationwide 2023	0.966	1	0.966	1.014	Yes
Nationwide 2024	0.951	0.9513	0.9997	0.9513	No
Nationwide 2025	0.974	1	0.974	1.1445	Yes
Santander UK 2022	1	1	1	1.0461	Yes
Santander UK 2023	0.9805	1	0.9805	1.0268	Yes
Santander UK 2024	0.9694	1	0.9694	1.1296	Yes
Santander UK 2025	0.9522	0.9711	0.9805	0.9711	No

Table 2 summarises the CRS, VRS, Scale Efficiency and Super-Efficiency for all 24 DMUs, with twelve of them achieving full VRS efficiency, while 7 achieve full CRS efficiency. Scale efficiency scores average 0.9726 across all DMUs with NatWest 2022 (0.9187) and HSBC 2024 (0.9291) recording the lowest values, where the small gap between their CRS and VRS scores confirms that underperformance is primarily driven by the scale size. For Barclays UK 2002 and HSBC 2022, both CRS and VRS fall below 1.0, indicating a combination of scale and managerial inefficiency. Despite this, both banks show clear improvement over the period, with NatWest reaching full efficiency by 2025 and Barclays UK rising from a CRS score of 0.8100 in 2022 to 0.9322 in 2025.

Table 3: Bank-level average efficiency scores (2022-2025)

Bank	Avg CRS	Avg VRS	Avg Scale	Avg Super-Efficiency	Years on Frontier
Santander UK	0.9755	0.9928	0.9826	1.0434	3
Lloyds	0.9878	0.9912	0.9965	1.0155	3
Nationwide	0.9728	0.9878	0.9849	1.0366	3
NatWest	0.9238	0.9486	0.973	0.9572	1
HSBC	0.8856	0.9169	0.9652	0.9745	1
Barclays UK	0.8799	0.9268	0.9504	0.9317	1

Table 4: Super-efficiency ranking for all frontier banks

Rank	DMU	Super-Efficiency	Outlier
1	HSBC 2023	1.2305	Yes
2	Nationwide 2025	1.1445	Yes
3	Santander UK 2024	1.1296	No
4	Lloyds 2025	1.0588	No
5	Santander UK 2022	1.0461	No
6	Nationwide 2022	1.0367	No
7	NatWest 2025	1.0345	No
8	Santander UK 2023	1.0268	No
9	Lloyds 2022	1.0249	No
10	Barclays UK 2025	1.0197	No
11	Nationwide 2023	1.014	No
12	Lloyds 2023	1.0137	No

Table 4 shows Lloyds and Santander UK have the strongest performance, with each reaching the efficiency frontier consistently in three years. Santander UK achieves the highest average VRS score of 0.9928, demonstrating strong managerial performance even at a smaller scale. Barclays UK and HSBC record the lowest average scores, reaching the frontier only once across the period.

In the case of super-efficiency, two DMUs have been identified as outliers, each exceeding the 1.1354 threshold, defined as 1.5 standard deviations over the mean. HSBC 2023 shows the highest super-efficiency score of 1.2305, influenced by exceptional net interest income in the peak UK interest rate period, while Nationwide 2025 records 1.1445 credited to the Virgin Money acquisition, inflating the balance sheet and revenue figures, and thus should be interpreted with caution as a structural outlier rather than an operational one.

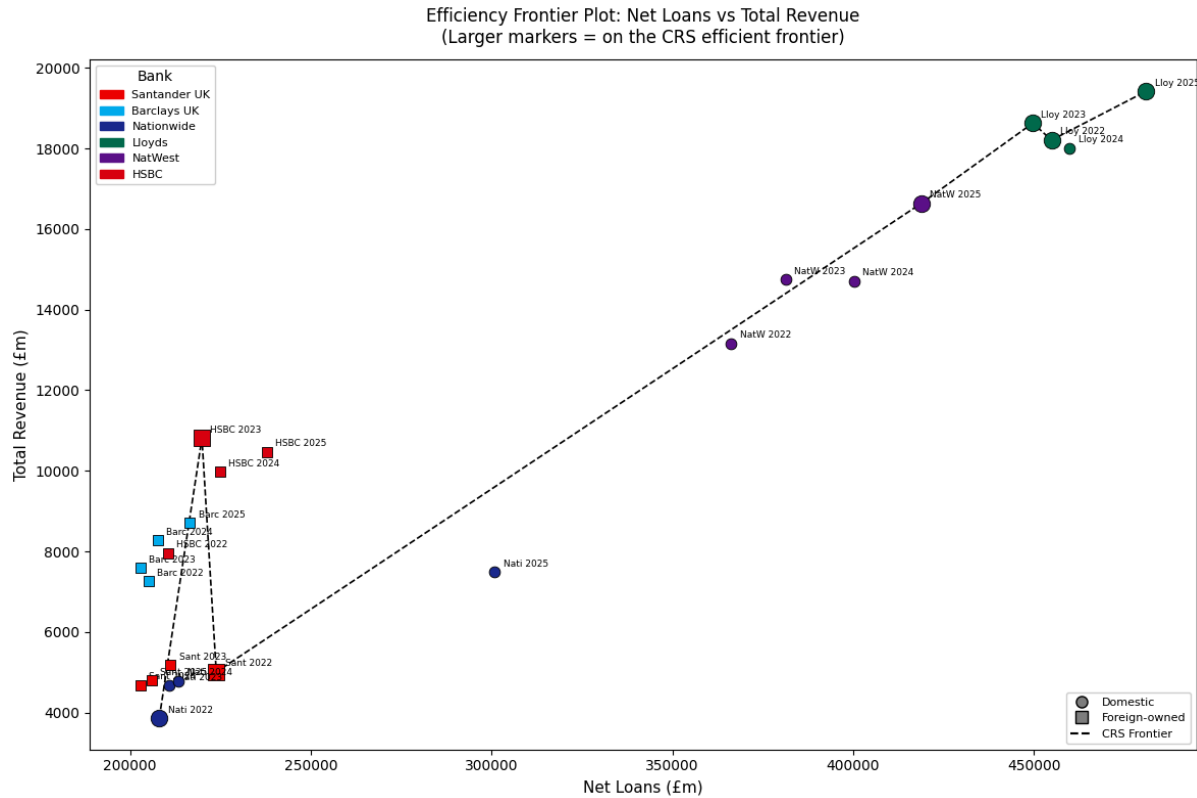


Figure 1: Efficiency frontier plot: Net Loans vs Total Revenue

Figure 1 confirms a clear structural separation between the larger domestic banks, Lloyds and NatWest, in the top right, and the smaller banks in the lower left. This confirms that the frontier is consistently defined by Lloyds, also defining the benchmark at a large scale.

Table 5: Group comparison - Domestic vs Foreign-owned banks

Group	Avg CRS	Avg VRS	Avg Scale
Domestic	0.9707	0.9786	0.9921
Foreign-owned	0.9322	0.967	0.9694

While Table 5 shows that domestic banks outperform foreign-owned banks across all three measures with the domestic group achieving an average VRS score of 0.9786 and a scale efficiency score of 0.9921, and the foreign-owned group achieve a VRS score of 0.9670 and a scale efficiency score of 0.9694, Mann-Whitney U test showed that the difference in VRS efficiency scores between the two groups is not statistically significant at the 5% level. The test returned a U statistic of 87.5 and a p-value of 0.355.

5. Business Insights

Translating the VRS scores into operational targets clarifies the magnitude of underperformance for each inefficient DMU. Under output-oriented DEA, the required output expansion is calculated as $(1/\text{VRS} - 1) \times 100\%$, indicating the proportional increase in net loans and total revenue each bank year would need to achieve, holding deposits and operating expenses constant, to reach the efficient frontier.

Table 6: Implied output expansion targets for the six most inefficient DMUs (VRS frontier)

DMU (Bank–Year)	VRS Score	Required Output Expansion	Managerial Implication
HSBC 2022	0.8133	+22.96%	Largest gap in the sample. Net loans and revenue would need to expand by roughly 23% to reach the VRS frontier given current deposits and operating spend, signalling material under-utilisation of inputs in the pre-rate-hike environment.
Barclays UK 2022	0.8305	+20.41%	Comparable to HSBC 2022 in scale of inefficiency; suggests the UK retail-banking segment was generating substantially less lending and revenue than peer benchmarks would imply.
HSBC 2024	0.8840	+13.12%	After reaching the frontier in 2023, HSBC reverted sharply. The 13% expansion target indicates that the 2023 performance was not embedded operationally and that cost growth outpaced revenue in the following year.
NatWest 2022	0.9056	+10.42%	Moderate inefficiency at the start of the period. NatWest closed this gap progressively, reaching the frontier by 2025, indicating an effective multi-year improvement trajectory.
Barclays UK 2023	0.9159	+9.18%	Continuation of Barclays UK's gradual improvement; the bank converged toward the frontier by 2025, suggesting that structural rather than transient factors drove the earlier underperformance.
NatWest 2023	0.9256	+8.04%	On a clear path to efficiency. Combined with NatWest 2022, the data suggest disciplined cost control alongside steady revenue growth as interest rates rose.

***Note.** Output expansion targets are derived from VRS efficiency scores reported in Table 2. Only the six DMUs with the largest gaps are shown; the remaining six inefficient DMUs each fall below a 6% expansion target.*

Finding 1: Foreign banks might show higher volatility: HSBC Case

A further finding from the analysis is that foreign-owned banks display greater efficiency volatility than their domestic peers, with HSBC providing the clearest illustration. HSBC recorded both the highest peak efficiency and the largest year-on-year variation in the sample: its CRS score moved from 0.7942 in 2022 to the frontier value of 1.0000 in 2023, fell back to 0.8213 in 2024, and partially recovered to 0.9270 in 2025, producing a range of 0.206 that exceeds any other DMU (NatWest 0.168; Barclays UK 0.122; Lloyds and Nationwide both below 0.05). Furthermore, HSBC 2023 DMU reached the CRS frontier and achieved the highest super-efficiency score in the entire sample at 1.231, ahead of Nationwide 2025 (1.145) and Santander UK 2024 (1.130). Taken together, these results indicate that HSBC was capable of materially outperforming its UK peers under favourable conditions, but was unable to sustain that position consistently. The fluctuation likely reflects yearly shifts in strategic allocation within the wider HSBC Group, changes in cost structure as the bank responded to the 2023 interest-rate environment, and inconsistencies in how the UK segment performed relative to the overall bank entity.

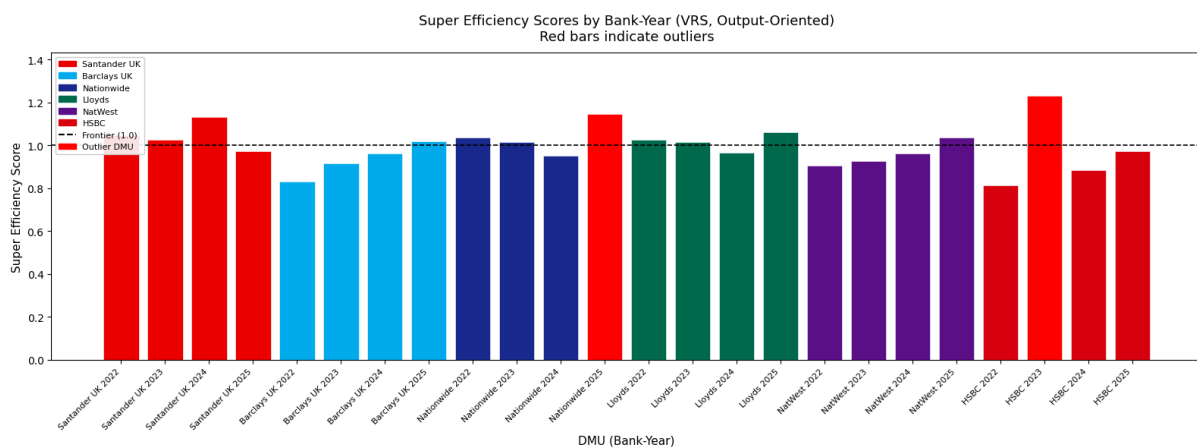


Figure 2: Super Efficiency Scores by Bank Year

Finding 2: Explaining the Banks Performances and Sources of Inefficiencies

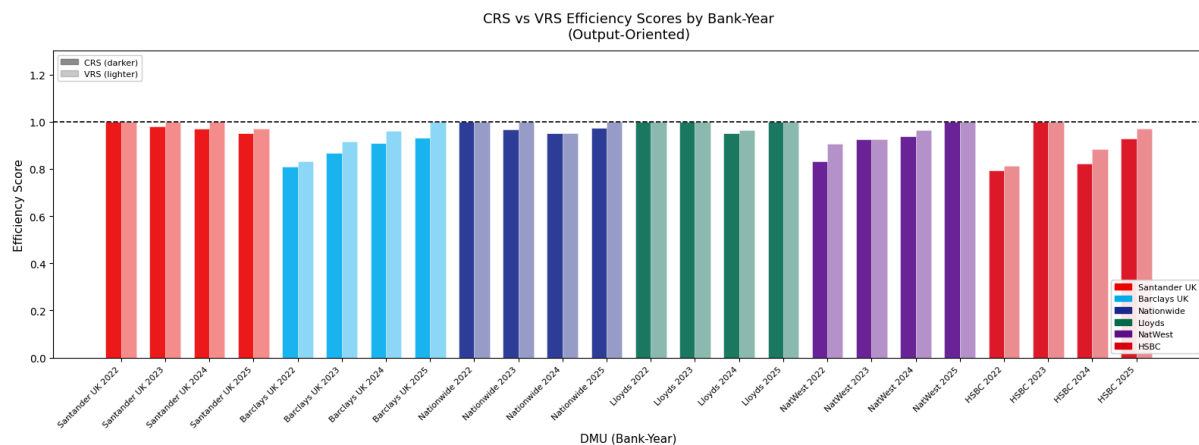


Figure 3: CRS and VRS Scores

By looking at Figure 3, we can classify banks into three main categories: industry leaders, progressing fellows, and unstable performance. The first group is represented by the Lloyds, Nationwide and Santander banks with the efficiencies on or close to the frontier. They also possess low degrees of fluctuation. In the second class, we may include Barclays and NatWest, with consistent and resilient performance improvements over 4 years, reaching CRS efficiency in 2025. Finally, the HSBC bank experienced the greatest score volatility, ranging from 0.7942 to 1 in the CRS model.

By referring to Table 2, we can observe that the inefficiency stems from an improper scale of operations. For example, Barclays 2025 appeared to be VRS efficient, while CRS and Scale were inefficient with a score of 0.9322. Hence, we have five DMUs with similar outcomes. On the other hand, we have no examples where DMU is a scale but not VRS efficient, which clearly supports the claim that underperformance comes from an unbalanced operational scope.

We can also emphasise that Lloyds appears to be the most operationally efficient bank as it operated at its Most Productive Scale Size three years out of four reviewed.

6. Conclusion

This study adopts the output-oriented Data Envelopment Analysis (DEA), combined with four models of CRS, VRS, Scale Efficiency and Super-Efficiency, to systematically evaluate the operating efficiency of six UK banks between 2022 and 2025, and derived two major

findings. First, the fluctuations in the efficiency performance of foreign banks are more obvious. Among them, HSBC recorded the highest super-efficiency score in the sample in 2023 (1.231), but this high level could not be continued in subsequent years. Second, while the count of frontier-efficient DMUs suggests scale considerations dominate, the magnitude of inefficiency among the lowest-ranked banks (Table 6) is primarily managerial rather than scale-based. Together, the findings give bank managers a benchmark for identifying performance gaps and offer investors a risk-adjusted view of bank performance that goes beyond traditional financial ratios.

7. Limitation

Beyond the lack of internal access to some in-house information, which could be a better estimate for input and output variables, there is also a notable limitation at the methodological level of this study, the high correlation between the two input variables. The correlation coefficient between total deposits and operating expenses reaches 0.953, reflecting multicollinearity. Although it does not violate the basic requirements of the DEA model, the overlap of information between input variables may still compress the input space and weaken the ability to distinguish the front-line DMUs, thus increasing the risk of misjudging the actual ineffective DMUs. (Adler and Yazhemsky, 2010). Earlier work by Ueda and Hoshiai (1997), Zhu (1998), and Adler and Golany (2001) establishes PCA-DEA as an extension when input redundancy is present. Adler and Yazhemsky (2010) show via Monte Carlo simulation that PCA-DEA produces more accurate efficiency classifications than partial-covariance variable reduction with high correlation among inputs. Applying this approach in a future iteration would allow the deposit–cost dimension to be summarised by a single component, sharpening the frontier and reducing the likelihood that 7 DMUs currently classified as CRS-efficient include false positives driven by correlated input information.

A second limitation concerns the interpretation of the comparison between domestically-owned and foreign-owned banks. Although domestic banks recorded higher average efficiency scores than foreign-owned banks (CRS: 0.971 vs 0.932; VRS: 0.979 vs 0.967, as shown in Figure 4), the gap is modest and may not be statistically meaningful given the limited sample of 24 DMUs drawn from only six banks. The small sample size constrains the statistical power of any group-level comparison, and the narrow distribution of efficiency scores further reduces the likelihood of detecting a significant difference, as reflected in the

Mann–Whitney U test result reported in Section 4. The observed efficiency gap between the two ownership groups should therefore be interpreted with caution and treated as indicative rather than conclusive. Future research drawing on a larger panel of UK banks, or extending the time horizon beyond the four years considered here, would help establish whether the apparent domestic advantage reflects a genuine structural pattern or is partly an artefact of sample-specific characteristics.

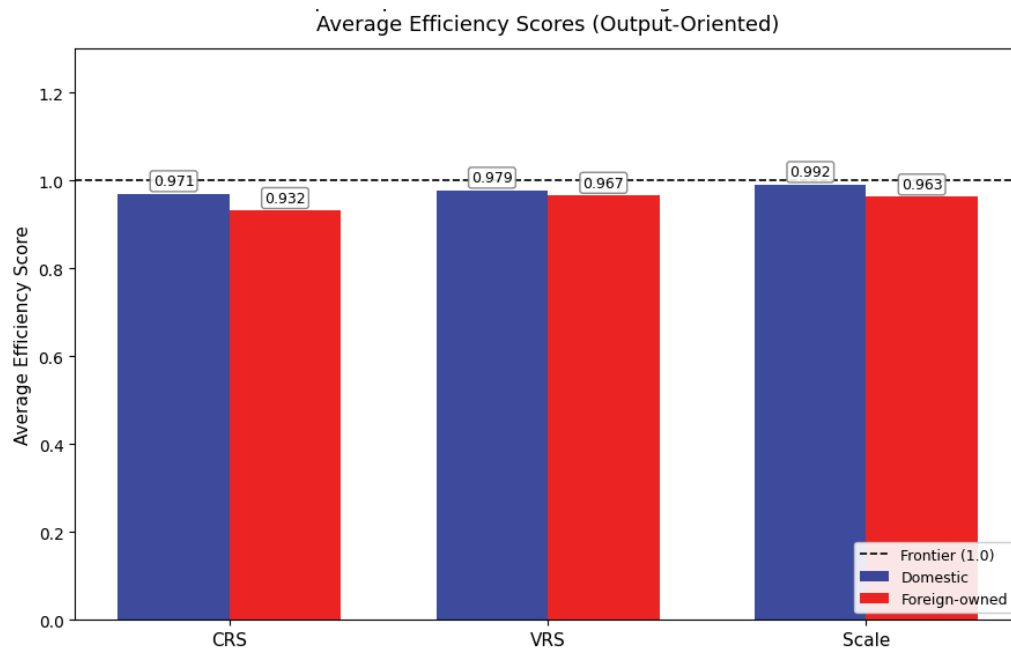


Figure 4. Group Comparison: Domestic vs Foreign-owned Banks

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Appendix

Appendix 1

Summary of all model notations

Symbol	Meaning
N	Number of DMUs (bank-years) in the sample
j	Index of the DMU being evaluated ($(j = 1, \dots, N)$)
j_0	The specific DMU currently being evaluated
m	Number of inputs in the DEA model ($(m = 2)$)
s	Number of outputs in the DEA model ($(s = 2)$)
X_{deposits}	Total deposits (£m) for DMU (j)
X_{opex}	Operating expenses before credit impairment (£m) for DMU (j)
Y_{loans}	Net loans (£m) for DMU (j)
Y_{revenue}	Total revenue (£m) for DMU (j)
λ_j	Peer intensity weight assigned to DMU (j)
ϕ	Output expansion factor (to be maximised)
Θ	Efficiency score for DMU (j_0), calculated as $(1/\phi)$
Θ^{CRS}	Overall technical efficiency score under CRS
Θ^{VRS}	Pure technical efficiency score under VRS
SE_j	Scale efficiency of DMU (j), calculated as $(\theta^{\text{CRS}}/\theta^{\text{VRS}})$

Appendix 2

Formulas

Output-Oriented CCR Model

$$\max \phi$$

Subject to:

$$\sum_{j=1}^N \lambda_j Y_{loans,j} \geq \phi Y_{loans,j_0}$$

$$\sum_{j=1}^N \lambda_j Y_{revenue,j} \geq \phi Y_{revenue,j_0}$$

$$\sum_{j=1}^N \lambda_j X_{deposits,j} \leq X_{deposits,j_0}$$

$$\sum_{j=1}^N \lambda_j X_{opex,j} \leq X_{opex,j_0}$$

$$\lambda_j \geq 0 \quad \forall j$$

$$\theta = \frac{1}{\phi}$$

Output-Oriented BCC Model

The BCC model extends the CCR formulation by adding the convexity constraint:

$$\sum_{j=1}^N \lambda_j = 1$$

Scale Efficiency

$$SE_j = \frac{\theta_j^{CRS}}{\theta_j^{VRS}}$$

Super-Efficiency Model

The super-efficiency model excludes the evaluated DMU j_0 from its own reference set:

$$\begin{aligned}\sum_{j=1, j \neq j_0}^N \lambda_j Y_{loans, j} &\geq \phi Y_{loans, j_0} \\ \sum_{j=1, j \neq j_0}^N \lambda_j Y_{revenue, j} &\geq \phi Y_{revenue, j_0} \\ \sum_{j=1, j \neq j_0}^N \lambda_j X_{deposits, j} &\leq X_{deposits, j_0} \\ \sum_{j=1, j \neq j_0}^N \lambda_j X_{opex, j} &\leq X_{opex, j_0} \\ \sum_{j=1, j \neq j_0}^N \lambda_j &= 1 \\ \lambda_j &\geq 0 \quad \forall j \neq j_0\end{aligned}$$

Appendix 3

Data Sources: All financial data were collected from published annual reports and regulatory filings for the financial years 2022 to 2025. The specific sources for each bank are as follows:

- Lloyds Banking Group - Annual Report and Accounts, available at: investors.lloydsbankinggroup.com
- NatWest Group - Annual Report and Accounts, available at: investors.natwestgroup.com
- Nationwide Building Society - Annual Report and Accounts, available at: nationwide.co.uk/about/corporate-information/annual-report
- HSBC UK Bank plc - UK segment data extracted from HSBC Holdings plc Annual Report and Accounts, available at: hsbc.com/investors/results-and-announcements/annual-report
- Barclays Bank UK plc - UK segment data extracted from Barclays plc Annual Report and Accounts, available at: <https://home.barclays/investor-relations/reports-and-events/annual-reports/>
- Santander UK plc - Annual Report and Accounts filed at Companies House, available at: companieshouse.gov.uk

Appendix 4

Generative AI Use Disclosure

In preparing this assignment, we used ChatGPT and Claude in the following ways:

- Used for brainstorming ideas, and ways to structure the DEA model.
- Learning and checking understanding of DEA and cross-efficiency concepts.
- Summarising Literature related to the paper and verifying references.
- Grammar check and structure verification.
- Whenever AI tools suggested explanations, code fragments or analytical ideas, we treated them as draft input only.

Appendix 5

Link to the code: [!\[\]\(5eb1325dfdc3f1cad8426726c0db51cd_img.jpg\) Banking DEA](#)

The model is implemented in Python using PuLP optimisation library.